

An Invitation to Computational Visual Humanities

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Advances in computing ranging from imaging, graphics, vision, to AI are transforming how humanists approach visual cultural heritage. This piece, co-written by computer scientists and humanists, argue that meaningful progress requires research agendas shaped by humanistic questions and expertise. Through examining four humanistic goals: recovery and conservation, interpretation and discovery, communication and access, and responsibility and ethics, we discuss how computation has contributed to visual humanities and highlight exciting opportunities ahead in the burgeoning field of computational visual humanities.

1 Introduction

Like virtually every other field, the humanities have been profoundly transformed by technologies. The field now known as *digital humanities* has its origins in the postwar tradition of humanities computing, often traced to Father Roberto Busa's *Index Thomisticus*, a computational concordance of Thomas Aquinas begun in the late 1940s. However, digital humanities as a broad scholarly enterprise emerged much later, particularly during the 2000s, when large-scale digitization initiatives, dedicated funding, and shared infrastructure transformed the field. During this period, the primary emphasis was on creating and maintaining digital collections, archives, and repositories that made cultural materials widely accessible through searchable digital repositories.

We are once again at the cusp of a new phase: the *computational humanities*. The transition from digitization to computation is profound. The focus is no longer just to digitize artifacts, but to computationally analyze, and to gain knowledge from, the artifacts. Advances in computer vision, graphics, imaging, and artificial intelligence (AI) are assisting scholars in unprecedented ways, aiding efforts to reconstruct missing information, understand creative processes, identify connections across disciplinary, linguistic, and disciplinary silos, as well as formulate new questions that would have been difficult or impossible to pursue through traditional methods alone.

Computational humanities offer the computing community not only impactful research problems and compelling motivations for developing new computational methods, but also intellectual interlocutors that challenge, refine, and expand the questions that computation can address. Critically, however, this requires research agendas driven not merely by technological capability, but by the quests humanists care most deeply about. Indeed, recent critiques of computational approaches applied to the humanities have lamented a tendency to prioritize algorithmic problem-solving in ways that flatten contextual nuance, recapitulate existing biases, and problematically bypass, rather than assist, humanist expertise [30].

The central thesis of this article, co-written by both humanists and computer scientists, is that computing researchers should understand how *humanists* conceive of computational humanities. While fostering meaningful engagement between computing and the humanities has received considerable attention from the humanities community [4, 20], there has been comparatively little discussion from the computing community. Addressing this gap is particularly important today, given that the rising ubiquity of AI has spurred a strong anti-computational backlash among skeptics disillusioned with valid concerns about the blunt application of digital methods not fine-tuned to humanities sensibilities [30].

Given the broad scope of humanities, this paper focuses on a particular branch of humanity — visual humanities — that are concerned primarily with visual arts and visual cultural heritage. In

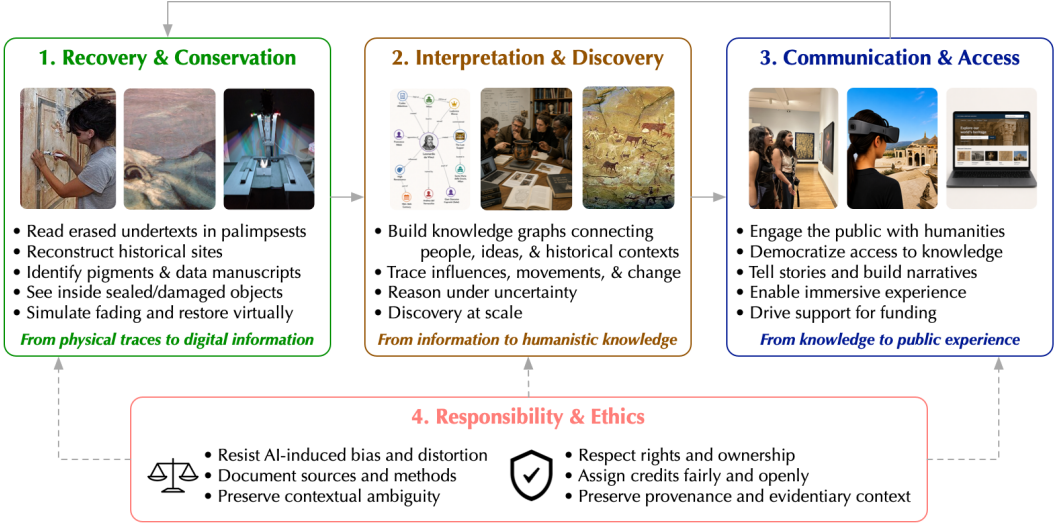


Fig. 1. A conceptual framework for the four main humanistic goals. The first three goals form a virtuous cycle: (1) **Recovery & Conservation** acquires and reconstructs historical evidence from physical artifacts; (2) **Interpretation & Discovery** transforms recovered information into humanistic knowledge through scholarly reasoning; and (3) **Communication & Access** disseminates knowledge through narratives, immersive experiences, and public engagement, which in turn motivate further recovery efforts and drive funding opportunities. Across all steps, (4) **Responsibility & Ethics** acts as a cross-cutting foundation that guides provenance, transparency, representation, ownership, interpretability, and fairness.

what follows, we discuss four main humanistic quests, and examine each in turn to outline ways in which computing has contributed, and conclude with exciting opportunities that lie ahead.

The humanities and cultural heritage communities are driven by four tightly connected goals that together motivate the development of computational methods and systems. Fig. 1 presents a framework that connects the four goals. The first three goals form a virtuous cycle, while the fourth goal intersects and influences all of the others.

- (1) **Recovery and Conservation:** Cultural artifacts frequently have hidden details invisible to the naked eye. An essential initial step is to document and retrieve the historical information embedded in these objects and places. Such recovery work is valuable in itself and also can inform conservators on how to best treat and restore damaged artifacts and build evidence so that it will continue to be available for in future generations.
- (2) **Interpretation and Discovery:** Recovering information is usually a step in the process, not the final goal. Once the raw material has been gathered, scholars interpret it to situate it within larger historical, cultural, and social frameworks. In the end, humanists aim to generate new insights and knowledge from the information acquired in the first step.
- (3) **Communication and Access:** Art and culture acquire value through engagement and shared understanding. Public engagement not only broadens access and appreciation but also helps build the societal support and funding needed to sustain future recovery, conservation, and discovery efforts.
- (4) **Responsibility and Ethics:** Cutting across all above is the broader goal of responsibility and ethics. This raises fundamental questions of validity, uncertainty, rights and ownership, and bias that must be addressed hand-in-hand with the technical questions.

2 Recovery and Conservation

The first goal, Recovery and Conservation, focuses on acquiring and reconstructing information from physical artifacts that has become inaccessible. Such information is critical for preservation and conservation. For instance, conservators can restore a painting using historically accurate materials only after identifying its original pigments; damage assessments and preservation planning at vulnerable cultural heritage sites often can be aided by the availability of detailed 3D models.

Information can become inaccessible for many reasons. For instance, medieval scribes commonly recycled parchments by scraping the older texts to write new ones; the remains of the undertexts are thus hidden behind the new, visible texts at the surface level. One of the most famous examples is the Archimedes Palimpsest, where the overtext is a medieval prayer book and the undertext contains two treaties of Archimedes that were thought to have been lost. Similar to palimpsests, paintings may contain pentimenti, where sketches or earlier compositions are concealed beneath later layers. Manuscripts may also be physically inaccessible due to damage. The Herculaneum papyri, for example, are heavily carbonized and remain unreadable by conventional means [26]; many manuscripts are too fragile to be fully opened without damage [18]. At a larger scale, entire cultural heritage sites can be partially destroyed by war or natural disasters.

Spectral Recovery. Information hidden from our eyes is not necessarily lost. The way a material interacts with light encodes a rich optical signature that can be used to identify, distinguish, and recover underlying materials. For instance, different layers in a palimpsest may be written with different inks, each characterized by a unique spectral signature that could enable their separation. Our eyes capture only the combined response of the ink mixture to light, obscuring the individual signatures of the underlying inks.

Advanced imaging modalities coupled with computational methods overcome this limitation by probing material properties through controlled light-matter interactions. A classic example is multispectral imaging (MSI) and hyperspectral imaging (HSI) [24] used to recover the undertext of the Archimedes Palimpsest [22]. The central idea, familiar from remote sensing, is that although a camera captures the combined response of mixed pigments, an *inverse problem* can be formulated to infer the underlying materials.

The inverse problem, however, is usually ill-posed, as it requires decoupling the material signatures and the composition of the materials, both of which are generally unknown. The problem becomes more challenging if the light-matter interactions cannot be expressed as a linear model because of, for instance, multiple scattering of the pigments, where non-linear models must be used [40, Chap. 15]. This is where computational methods come in. These methods incorporate physical and statistical priors, which help constrain the space of feasible solutions and recover physically meaningful material signatures and abundance maps [7].

Beyond MSI and HSI, a variety of other spectroscopic imaging modalities have been also used in cultural heritage [31], such as X-Ray Radiograph, X-Ray Fluorescence (XRF), Ultraviolet Fluorescence, Raman spectroscopy, and Fourier transform infrared (FTIR) spectroscopy. Each modality probes a different aspect of light-matter interactions. In many cases, therefore, multiple modalities are combined to increase the number of observations and provide additional constraints on the recovery problem. Regardless of the sensing modality, the role of computation is central: imaging systems acquire raw measurements, and computational methods solve the inverse problems to recover the material, structural, and historical information encoded within the data.

Geometric and Volumetric Recovery. In addition to recovering the spectral signatures of materials, art historians and conservators are often interested in the geometric and volumetric structure of cultural heritage artifacts. Addressing this challenge has drawn upon a rich body of work in computer vision and computer graphics, coupled with advanced imaging modalities. For

instance, X-ray Computed Tomography (XCT) and Optical Coherence Tomography (OCT) can provide micron-level spatial resolution and have been used in recovering surface and subsurface geometries in historical documents such as the En-Gedi scroll (the oldest Pentateuchal scroll in Hebrew outside of the Dead Sea Scrolls) and the Herculaneum Papyri [3].

While XCT and OCT offer micron-level spatial resolution, both require elaborate hardware setups. Researchers have thus explored purely photometric approaches that use conventional cameras and rely more heavily on computational methods. Photometric stereo uses a fixed camera to capture images under varying illumination directions and solves an inverse problem to estimate surface normals and recover fine-scale geometry [37]. Building on photometric stereo, Reflectance Transformation Imaging (RTI) incorporates polynomial texture mapping [25] to enable virtual relighting of artifacts, allowing conservators and historians to inspect subtle surface features such as worn inscriptions, brushstrokes, tool marks, and seal impressions.

Different from photometric stereo where the camera is fixed and the illumination varies, another, arguably more popular, form of photometric 3D reconstruction does the opposite, where the camera moves to capture images under different poses but the illumination remains unchanged. This is usually called structured from motion (SfM) or photogrammetry, and has been widely used in reconstructing large 3D scenes in cultural heritage. Using SfM, Snavely et al. [34] builds a 3D model for the Colosseum in Rome, and Al Khalil [1] reconstructs the medieval Safita Tower in Syria.

Yet another commonly used method is based on triangulation, where only one fixed illumination source and one fixed camera are involved. Critically, the spatial positioning of the source and the camera are pre-calibrated to enable triangulation. The Digital Michelangelo project uses a laser triangulation system to create 3D models of many Michelangelo's sculptures, including the *David* [23]. The IBM Pietà project uses a structured-light triangulation system to create a 3D model for Michelangelo's *Florentine Pietà* [6].

Finally, geometry information can also be acquired using a time-of-flight laser/LiDAR scanner, often at a sub-millimeter spatial resolution. The Digital Elmina project uses a portable LiDAR scanner (coupled with photogrammetry) to capture 3D models of the Elmina Castle in Ghana, a UNESCO World Heritage site, which can then later be explored in Virtual Reality [39]. Likewise, the restoration of Notre-Dame Cathedral following the 2019 fire was aided by detailed laser-scanned 3D models acquired prior to the disaster.

3 Interpretation and Discovery

Recovering information is the means, not the end. Humanists are ultimately concerned with *hermeneutics*, i.e., transforming information into new *knowledge* about people, societies, cultures, and the human experience. As Hannah Ringler puts it: "As computational humanists, if we want to contribute to humanistic inquiry, computational analysis is only useful for us insofar as it is paired with a careful interpretation of data that aligns with the humanistic tradition. In other words, our computation needs a hermeneutics." [20, Chap. 1].

How Are Arts Created? Humans have long been fascinated not only by what past artists produced, but also by how they produced it. Conservators at National Gallery of Art use XRF and HSI to reconstruct the hidden layers beneath the visible paint surface in two of Vermeer's famous paintings, *Woman Holding a Balance* and *A Lady Writing* [14], which reveals new knowledge about Vermeer's artistic practice. Beneath the carefully finished surfaces in both paintings, Vermeer often worked rapidly during the underpainting stage, using broad brushstrokes and coarsely ground pigments. This not only challenged the longstanding view in art history that Vermeer worked slowly and meticulously throughout the entire painting process, but also showed how the scarcely apparent underlayers contributed to Vermeer's finished works.

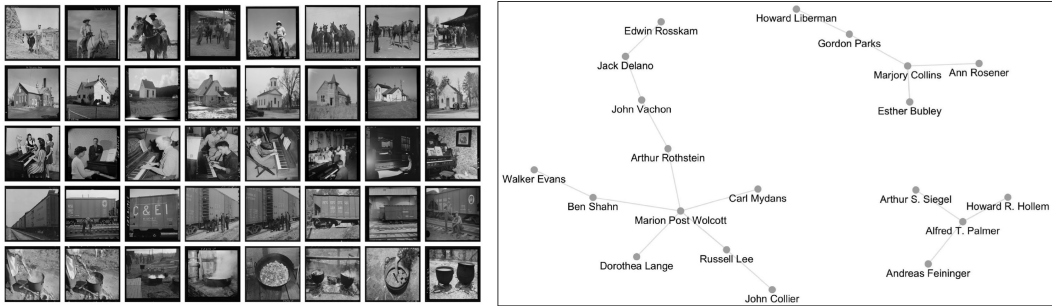


Fig. 2. Distant viewing of the FSA-OWI photographic archive (adapted from Arnold and Tilton [4]). Left: Example groups (one per row) of visually similar photographs retrieved using latent representations from the InceptionV3 neural network. Right: A network of staff photographers where each photographer is connected to the photographer whose images are most visually similar, revealing patterns of influences between photographers.

Defrasne et al. [11] uses HSI in the infrared (IR) regime to assist the detection of numerous examples of otherwise invisible rock art in the Otello rock shelter in southern France. The HSI data help classify pigments, group figures in rock art, and distinguish and characterize weathering crusts, contributing to our understanding of the site’s taphonomy.

Computational methods have also helped in weighing into art historic analyses and debates. For instance, David Stork and colleagues use multi-view computer vision techniques to show that the perspective is far from perfect in a painting of Jan van Eyck’s, an example used by some, including David Hockney, to argue that some Western artists used optics to assist their drawing [35]. Leonardo Impett argued for how digital art history could use algorithms to track gestures and poses across paintings, enabling researchers to identify patterns across large bodies of art [19].

The power of computational methods has also led to new theories and methods, such as Arnold and Tilton’s concept of *distant viewing* — a method for large scale analysis of visual culture that exceeds human ability [4]. Their analysis of roughly 170,000 photographs in the Farm Security Administration–Office of War Information (FSA-OWI) archive illustrates this potential. They use latent representations from the InceptionV3 neural network to retrieve visually similar images across the collection, with several example groups shown in Fig. 2 (left), and then construct a network of staff photographers where each photographer is connected to the photographer whose work is most visually similar to their own, shown in Fig. 2 (right). Such analyses can reveal patterns that prompt new questions about artistic influence, shared practice, and the broader structure of visual culture. More broadly, Arnold and Tilton have applied distant viewing to tens of thousands of artworks to examine how institutions such as museums shape what is recognized as art.

How Do Humans Record and Transmit Knowledge? Another fundamental humanistic quest is to understand how societies record, communicate, and preserve knowledge across generations. A prime example is the study of cuneiform, one of humanity’s earliest writing systems. Analysis of cuneiform tablets has enabled scholars to show the transmission of texts between the Old Babylonian period and the first millennium [36] and to understand the physical environment of education and the scribal curriculum in Mesopotamia [32].

A wide range of computational methods have been developed to assist in recognizing and analyzing cuneiform scripts, including 3D acquisition and modeling, relighting, segmentation, and pattern mining. The Palaeographicum initiative¹ provides AI-based tools for extracting cuneiform

¹<https://smaw.de.dariah.eu/palaeographicum/>.

signs from photographs of clay tablets. By lowering the barriers to transcription and enabling large-scale analyses, computational methods open new avenues of inquiry. How did scribal practices evolve across regions and centuries? How did changes in political power and state institutions reshape writing conventions and documentary practices?

How Did Ordinary People Experience Religion in Late Medieval Europe? Books of Hours are personal prayer books used by laypeople in the late medieval period and are often described as “medieval bestsellers.” [9]. Many copies were customized for their owners. Owing to their widespread circulation and remarkable diversity, Books of Hours constitute one of the richest sources for studying the religious, social, and cultural life of late medieval Europe.

Yet precisely because of their abundance, systematically examining and comparing them is beyond the reach of any individual scholar. The DESMM project [9] and the HOARE project [8] approach this challenge by developing computer vision and machine learning techniques to analyze Books of Hours at scale, automatically segmenting pages into text and illustration regions, classifying page layouts, and comparing features across manuscripts.

These methods make it possible to surface a larger sample of objects for domain experts to further investigate. For instance, scholars consistently identified vernacular passages (e.g., in English) alongside the primary Latin text, which invites broader questions about late medieval Europe that historians can further probe: How were devotional texts adapted for local audiences? How bilingual were medieval readers? And how did literacy spread beyond clerical elites?

4 Communication and Access

A central goal of the humanities research is to help individuals understand themselves and their place within broader worlds. Yet much humanistic knowledge remains inaccessible to non-specialists. Computing technologies have helped bridge this gap, offering tools that domain experts can use to transform humanistic knowledge into experiences that are engaging, educational, and accessible.

Immersive Story Telling. One promising direction is immersive storytelling. Many aspects of human culture are inherently *experiential*: textbooks and photographs can describe a medieval city, but they often struggle to convey scale, spatial relationships, atmosphere, and the lived experience of being there. By combining 3D reconstruction, virtual and augmented reality, and real-time interactions, immersive storytelling enables people to engage with cultural heritage in ways that are more intuitive, memorable, and emotionally resonant.

A compelling example is Virtual Angkor², which combines archaeological evidence, 3D LiDAR surveys, and historical scholarship to build an interactive 3D platform of the Angkor, the world’s largest preindustrial settlement complex. Using the platform, researchers populate the virtual environment with many autonomous agents, each representing a resident going about daily activities. They then simulate a 24-hour cycle at Angkor Wat, allowing the public to follow individual agents to experience what life might have been like in a preindustrial city. Users can also explore Angkor Wat through Virtual Reality.

Education and Pedagogy. Immersive experiences can also facilitate pedagogy. Smithsonian Voyager³ is an interactive platform from the Smithsonian Institute, and has been particularly popular for pedagogical uses. Educators can attach annotations to specific regions of a 3D model, enrich them with multi-modal data, and organize them into guided tours that lead students through a curated sequence of interpretations. A recent humanities teaching tutorial from Carleton College

²<https://www.virtualangkor.com/>.

³<https://3d.si.edu/voyager-story-standalone>.

illustrates how students use Voyager Story to build annotated explorations of 3D models and communicate historical interpretations through interactive tours⁴.

Gamification, or learning-oriented games, offers another powerful avenue for communicating humanistic knowledge to broad audiences [12]. The Iñupiaq Native American tribe of Alaska published *Never Alone (Kisima Ingitchuna)*⁵, a video game where players get rewarded by collecting “cultural insights”: unlocking short documentary videos in which Iñupiat elders, storytellers, and community members discuss oral traditions, environmental knowledge, and cultural practices. It has been an extremely successful tool in teaching the Iñupiaq to the public and support cultural preservation and make underrepresented voices accessible to global audiences.

5 Responsibility and Ethics

While computational methods have the potential to help make significantly advances in humanities, they must be used responsibly and ethically throughout each stage of the process [13].

Validity and (False Sense of) Certainty. Generative AI technologies offer powerful new tools for recovering and reconstructing missing or inaccessible information, but they also raise a fundamental question: where does reconstruction end and speculation begin? For example, Moral-Andrés et al. [27] explored using generative AI to reconstruct missing portions of ancient Roman mosaics. While many of the results are visually compelling, some reconstructions introduce patterns and motifs that are inconsistent with the surviving evidence. As the *apparent* quality of generative models continue to improve, such discrepancies may become increasingly difficult to detect.

Worse, as AI systems produce seemingly compelling outputs, they can also create the illusion that historical facts, meanings, and cultural values can be determined with certainty. In reality, these questions are often contested and subject to ongoing interpretation by scholars, stakeholders, and communities. The challenge is particularly acute in cultural heritage, where it may be impossible to definitively decide among competing iconographic interpretations of the same artwork due to the limitations of the preserved evidence, or where many reconstructions necessarily contain conjectural elements because the surviving archaeological and historical evidence is incomplete.

Humanists have long recognized that highly realistic reconstructions can create a false sense of certainty. The London Charter and the Seville Principles⁶ therefore emphasize the importance of documenting evidence, assumptions, and uncertainty so that viewers can distinguish attested facts from scholarly hypotheses. As Nygren and Drimmer [30] argue, one of the central axioms for applying AI to art history is that “machine learning can change how we see the historical record, but it cannot complete the historical record.”

Fundamentally, for AI to be helpful, we require architectures that prioritize source transparency and data grounding over the tendency to flatten interpretive differences into a singular, highly polished output. For instance, coupling Retrieval-Augmented Generation (RAG) alongside specialized models trained exclusively on domain-specific cultural heritage corpora could offer a promising framework for surfacing “noisy” data and preserving conflicting historical hypotheses.

Rights, Ownership, and Benefits. Digitization and AI can dramatically expand access to cultural heritage, but they also raise questions of rights to access, ownership, and benefits: who has access to cultural heritage artifacts and digital byproducts, who owns them, and ultimately who benefits from the broader dissemination of cultural knowledge?

Recent research is making visible persisting inequities tied to who can access and use existing digital resources [2, 10]. As an example, many museums have digitized Native American artifacts

⁴<https://hh2025f.amason.sites.carleton.edu/blog/how-to-use-smithsonians-voyager-story/>.

⁵<https://www.neveralongame.com/>.

⁶<http://sevilleprinciples.com/>.

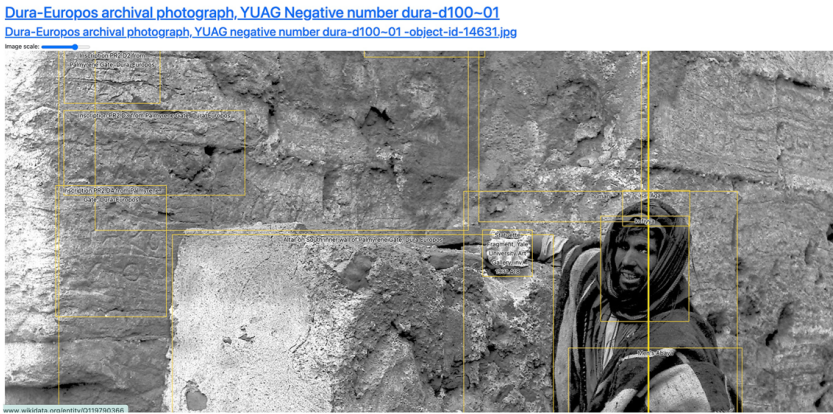


Fig. 3. Collaborative annotations in Wikidata for a Dura-Europos archival photograph, demonstrating the IDEA model of richer, multilingual, and more inclusive metadata; adapted from Alkhalaf et al. [2].

to increase access and support research; contrary to the common assumption that greater access is always beneficial, however, indigenous scholars have argued that some cultural materials should be shared according to community-defined protocols that specify who may access, use, and interpret them. Initiatives such as Mukurtu⁷ and the Traditional Knowledge Labels project⁸ were developed in response to these concerns. For instance, Mukurtu, as a content management system, empowers communities to share their digital cultural heritage “in their own way, on their own terms.”

The IDEA project, meanwhile, has pioneered a different approach; it uses the collaborative space of Wikidata to digitally upskill the global heritage peers and empower them to reshape exclusionary metadata and interpretive narratives tied to foreign-run excavations at the Syrian archaeological site of Dura-Europos [2, 10]. The original institutional record describes the archaeological monument but omits other historically meaningful information visible in the image, including the Syrian worker prominently depicted in the foreground. Through IDEA, collaborators enriched the corresponding Wikidata record with additional multilingual and locally grounded annotations, shown in Fig. 3, making previously overlooked people, places, and histories more visible and discoverable.

Bias. Will some cultures be permanently excluded in the future because they are insufficiently represented in today’s digital record? The concern is not hypothetical, given that the quality of AI depends on the volume and quality of the data. Human bias has plagued traditional art scholarships, and unscrupulous use of (generative) AI is already amplifying and re-entrenching such biases.

Many cultural heritage collections remain undigitized, while others are concentrated in well-funded institutions in North America and Europe. A growing body of scholarship rooted in Data Feminism [13] and critical archival practice is beginning to shed light on the over-representation of voices, perspectives, and collections from the Global North in the current digital cultural heritage ecosystem. UNESCO has repeatedly noted that many regions in the Global South face significant barriers to digitization, including limited funding, infrastructure, technical expertise, and linguistic resources⁹. Generations of art historians have also focused disproportionately on a small canon of predominantly male European artists, while women and non-Western artistic traditions have historically received less scholarly attention [29]. As a result, large-scale datasets used to train AI systems often provide substantially greater coverage of certain languages, artistic traditions, historical periods, and cultural perspectives than others.

⁷<https://mukurtu.org/>.

⁸<https://localcontexts.org/>.

⁹<https://www.unesco.org/en/articles/cutting-edge-protecting-and-preserving-cultural-diversity-digital-era>.

6 Outlook

The intersection of computing and cultural heritage is undoubtedly a burgeoning field. The number of publications appearing in the *ACM Journal on Computing and Cultural Heritage* (JOCCH) has increased more than sevenfold since the journal’s inaugural issue in 2008. As of 2024, Clarivate places JOCCH above the 95th percentile in the interdisciplinary category in both Computer Science and Humanities. Many schools now offer programs at the intersection of computing and cultural heritage [28], such as the Visual Heritage Ph.D. program in Austria¹⁰.

The success stories described above serve as powerful reminders of how far the field has come in advancing longstanding humanistic quests. Yet they likely represent only the beginning of what is possible. Recent advances in virtually all areas of computing are rapidly expanding the range of questions that scholars can ask and answer. In what follows, we highlight several particularly exciting opportunities and, importantly, challenges we must address.

Citizen Conservators. Once confined to specialized labs, advanced imaging hardware and computational methods are becoming increasingly ubiquitous and accessible. This democratization opens up the possibility of “citizen conservators,” much as platforms such as Zooniverse¹¹ and iNaturalist¹² have enabled citizen scientists in other scientific fields. For example, many modern smartphones now incorporate LiDAR sensors, making 3D capturing and modeling accessible to a broad public. Radiance-field-based rendering has dramatically lowered the barrier to high-quality photometric 3D reconstruction. Meanwhile, advances in physically based differentiable rendering are improving our ability to recover intrinsic material properties, including spectral reflectance [38].

The proliferation of 3D modeling also raises serious ethical questions. Where will all of this data be stored, who owns it, and how do we ensure that all stakeholders can access it? How can its authenticity be verified? How do we prevent bad actors from polluting digital archives with fabricated or manipulated historical records? Addressing these challenges will require the computing and humanities communities to work together to rethink, as discussed in Sec. 5, issues of validation, provenance, rights, ownership, and bias in the era of computational humanities.

Humanists-AI Collaboration. A recent white paper [17] from UK’s Alan Turing Institute calls for “developing frameworks for sophisticated, collaborative human-AI systems that strengthen our collective intelligence and enhance rather than replace human capabilities in complex decision-making.” A key risk that AI poses is reductionism, where complex cultural meaning and contextual nuance are reduced to a simple numerical prediction. This risk is particularly pressing in computational humanities, because many objects of humanistic inquiry are naturally incomplete and ambiguous. They can also be examined in many different ways through, for instance, topic modeling, sentiment analysis, and virtual relighting, each potentially revealing different aspects of an object that might not always be compatible. This multiplicity, or “computational parallax” as Crystal Hall calls it [20, Chap. 4], is a feature not a bug. What this calls for, however, are tools that can assist humanists to navigate the heterogeneous information and to reason under uncertainty. Reflecting on the Mapping the Republic of Letters project¹³, Edelstein et al. [15] ask: “Where are the tools to support humanistic inquiry into multidimensional, heterogeneous, and incomplete datasets?” They argue that they “require different tools that can accommodate ambiguities, paradoxes, and contingencies” in order to effectively reason about humanistic data.

Another risk, as discussed in Sec. 5, is the bias introduced or reinforced by AI. Recent work has highlighted the need for multi-stakeholder collaboration to address bias in heritage collections [13].

¹⁰<https://visual-heritage.ustp.at/>.

¹¹<https://www.zooniverse.org/>.

¹²<https://www.inaturalist.org/>.

¹³<http://republicofletters.stanford.edu/>.

In practice, however, language differences, different levels of comfort with technology, and limited ability to travel often present significant barriers [2, 10]. How might we refocus our use of computational tools (including, but not limited to AI) to bridge such real-world barriers?

Knowledge-Grounded, Generative Cultural Experiences. One promising direction of leveraging generative AI is not to use them to generate new historical records, but rather to generate new *experiences* grounded in verified historical knowledge. For example, could a generative system draw upon an expert-curated knowledge graph to answer natural-language questions in a range of global languages? Building on ideas explored in Virtual Angkor, generative AI could also populate virtual historical environments with autonomous agents that interact with one another and their surroundings according to historically informed constraints. Visitors could follow these agents through an ancient city, observing daily routines, social interactions, and economic activities, and thereby gain a richer understanding of what life may have been like in the past.

How Should Humanists and Computer Scientists Collaborate? Ultimately, the future of computational humanities will depend not only on better algorithms, larger datasets, or more powerful AI systems, but also on our ability to foster meaningful collaboration between the two communities. Take 3D reconstruction as an example. While there have been ample examples demonstrating the value of computational methods in assisting the conservation of historical information, the debate over the 3D reconstructions of the triumphal arch from Palmyra destroyed by ISIS in 2015 [21] stands as a pointed reminder of the need to ensure output fidelity, and of the danger of prioritizing data collection without ensuring shared benefits of the digital products, especially in contexts marked by longstanding power differentials among stakeholding groups.

There are also structural challenges facing the field of computational humanities. For instance, funding and evaluation mechanisms often favor disciplinary work over long-term interdisciplinary collaborations. For computing researchers, not every humanistic question gives rise to a new computational problem; many important humanistic discoveries may stem from the thoughtful application of existing methods to new domains and datasets. Such work should be recognized and rewarded, rather than dismissed or discouraged by the evaluation systems. As the authors of *Computational Humanities* emphatically ask: “can they fit alternative ways of working into the institutional structures that they must navigate to maintain their careers and their entire fields? Which new structures do we need to build and for whom?” [20].

At the same time, the next generation of breakthroughs is likely to emerge from where computational techniques advance humanistic inquiries, which, in turn, advance computational techniques. For many years the National Research Council Canada used cultural heritage to drive and demonstrate the development of laser scanners [16], which lead to better cultural heritage imaging. The massive point clouds of the Digital Michelangelo Project spurred the development of point-based rendering algorithms [33], while the need to reconstruct meshes from millions of points in the IBM Pietà Project led to the Ball-Pivoting surface reconstruction algorithm [5]. These techniques, along with later developments, lower the barrier of rendering large-scale cultural heritage datasets, further accelerating humanities research.

7 Conclusion

Computational visual humanities invites us to rethink both what computing can do and what it should pursue. In the IBM Pietà project, the demands of art historians pushed computing researchers to develop capabilities that engineering or scientific users perhaps would have not pushed. The challenges in the AI era, such as preserving ambiguity, tracing interpretations to evidence, and supporting multiple perspectives, can similarly drive the next advances in computing. Pursuing these challenges together gives computing an ambitious purpose: to deepen human understanding of, and strengthening our ability to question, interpret, and care for, the world we inherit.

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